

Fascia-oriented strength training in a conventional gym environment

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Introduction

Previous chapters described fascia-oriented training concepts, which do not use any external tools, or they use mobile tools such as small weights, or take advantage of stationary equipment—like the Pilates reformer—which is specially designed to allow for a rich variety of joint motions. However, for many of our clients, modern fitness gyms, with their stationary muscle-strengthening machines, have become a regular or at least easily accessible part of their lifestyles.

Muscle strength-oriented training offers indeed many health benefits that cannot be equally replaced by cardiovascular training or by fascia-oriented training. For example, several myokines—cellular messenger substances produced by muscle cells—have been described, which are upregulated after muscular strength training and exert beneficial anti-aging effects, not only on the trained muscles themselves, but also on most other organs and tissues in the human body, including our brains (Pedersen, 2019). Fascia training should therefore not be recommended as an alternative but rather as a useful complement to muscular strength training. This chapter describes how fascia-oriented training can be conducted in a conventional gym environment.

Panther versus bull

A shift in attitude is recommended when switching to a fascia-oriented movement style in a gym.

This can be nicely illustrated by the movement style of a panther, versus that of a bull (Schleip and Buschmann, 2016). A bull often increases its power by making the body more condensed—i.e. shorter, more compressed and more rounded—to increase the power behind its pushing force. This is also how some people push the resistance handles away in a conventional gym exercise, and it seems to work well for that purpose. What we advocate for a fascia-oriented workout style is to work with less resistance than in the bull style, but to emphasize a more graceful and elongated orchestration, maybe similar to the movements of an elegant panther. As a basic orientation we recommend starting with 50–60% of the usual resistance load, compared with a bull-type application style, although that percentage may need to be adjusted to a significant degree downwards or upwards for a given individual and situation.

Unless a more specific application purpose is chosen (as described later), we recommend working in two to three exercise sets per machine, with the following steps per set:

- Concentric phase 2–3 s
- Three to five mini-bounces around end point
- Eccentric phase 2–3 s
- Three to five mini-bounces around end point.

These should be performed with 90-s breaks between sets. The resistance should allow 10 to 15 cycles over the full range. If it allows more or

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only less than that then it should be adjusted. A specific myofascial tissue region should not be trained in this manner more than once or twice per week (with gaps of at least 2–3 days in between) to allow for sufficient recovery and adaptation. A time of 3 months can be expected before a visible and palpable tissue improvement can be noticed with this training style, based on the relatively slow remodeling times of dense collagenous tissues (see Chapter 1).

Principles of Panther training

The principles of Panther training are outlined below. In a basic application, the exercise starts with a preparatory counter-movement in the opposite direction to that anticipated for the main resistance movement. For example, in the example shown in Figure 43.1, the individual starts by extending their hands away from the chest and slightly backwards.

Panther training

Basic principles:

1. Preparatory counter movement
2. Proximal initiation
3. Directional variations
4. Mini bounces.

Advanced principles:

1. Tensegral pre-tension
2. Final exhaustion at end range
3. Embodiment.

Next, the main movement, against the resistance of the machine, is not initiated at the distal contact point with the equipment (e.g. the hands holding a bar), but rather at a much more proximal body part. In the example shown in Figure 43.1, this means that a small reaching-forward motion of the lower chest region precedes the subsequent forward motion of the arms. The combination of



Figure 43.1

For better loading of the fascial tissues, it is recommended to start the exercise by a preparatory counter-movement (extending the hands away from the chest and slightly backwards) before the main forward motion is initiated with a proximal body part, in this example with the lower chest.

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these first two principles tends to increase the tensile load on the fascial components during the subsequent main motion. For the main motion, the inclusion of different directional variations is recommended, such as shown in Figure 43.2. In addition, the inclusion of three to five quick elastic mini-bounces at both the long-range as well as the short-range end positions is recommended. A brief pause of a few split seconds after the last bounce ensures that the subsequent slow resistance motion includes the end point position at which the mini bounces occurred.

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Figure 43.2

For conventional muscle fiber training, exact repetitions along the same loading vector are usually recommended for optimal training effectiveness. By contrast, the fascia-oriented Panther training suggests exploring different loading vectors. This way many different tissue areas within larger myofascial sheets can be reached within the same exercise set.

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When familiar with these basic principles, an inclusion of the following advanced principles can be explored. Before starting an exercise, a brief moment (like one breathing cycle) should be devoted to slightly increasing the fascial pre-tension via a multidirectional expansion of the whole body in an integrated manner, which is described as “tensegral movement” by Dr. Daniele-Claude Martin (2016). In the example shown in Figure 43.1, this could mean a slight simultaneous extension of the whole body upwards and downwards together with subtle widening of the chest and arms.

Another added principle includes continuing the resistance movement at the final long-range position (in case this is possible) over shorter distances, until even that is no longer possible. In conventional muscle strength workouts, one is advised to repeat the basic movement as long as possible but to stop once a “clean” movement over the full range of motion is no longer possible. By contrast, in our fascia-oriented workout style, the trainee then continues the resistance motion over a much smaller distance (“squeezing the last drop out”) until the body exhaustion prevents

even this small motion. This serves to enhance an increasing range-of-motion remodeling effect, as will be shown later. It should not be included for individuals who have been classified as hypermobile after undertaking the Beighton test (see Chapters 7 and 19).

Finally, each set can be finished with a brief moment of open attention, in which the trainee pays curious attention with a mindful attitude towards any positive or interesting proprioceptive, interoceptive or emotional after-effects in themselves. While in a conventional muscle gym exercise it is not uncommon to see the trainee fall into a “slump position” for a brief moment when being exhausted at their last repetition, an individual engaged in a Panther workout will sit (or stand) quietly for a brief moment in a continued tensegral posture, and may feel tiny expansional body changes or increased vitality perceptions in the body, similar to listening to the after-effects of a stone thrown into a quite pond (Figure 43.3). Only after such a brief “embodiment moment” does the trainee then continue with the next set or with the next exercise.

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Figure 43.3

In the advanced version of our recommended Panther training, the trainee completes each set with a brief “embodiment” moment, in which any immediate after-effects in terms of altered body sensations or emotional changes are greeted with an attitude of open-minded curious attention.

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Specific guidelines for different purposes

Hypomobile type

If you have been classified as a general hypomobile type in the respective Viking Test (Chapter 16) then it is suggested that you perform only the first move of each set over the full available range of motion, and that you continue to conduct all subsequent moves over the last quarter of the available range of motion in the long stretched position for the primarily loaded muscles. It is also recommended that you conduct the mini-bounces in the long stretched position only (and not in the opposite position, in which the loaded muscle fibers are shortened), and that you apply the principle “Final exhaustion at end range” with enhanced emphasis. And, as an additional booster, we recommend conducting the eccentric phase—i.e. when the machine lengthens your muscle fibers against your resistance—twice as slowly compared with the concentric phases.

These recommendations are based on the research of Aquino et al. (2009), who showed that muscular resistance training in a stretched position induced a more powerful long-term increase in hip flexion mobility than a conventional stretching application. In addition, new insights about the role of the intramuscular titin fiber showed that this unique fiber plays a key role as a “third contractile element” (in addition to actin and myosin), and also as a very important mechanosensor, the signaling of which takes priority over the signaling from other mechanoreceptors. In particular, titin signaling tends to induce a long-term increase in muscle fiber length. Such titin signaling only happens when muscle fibers are either contracting at long lengths (independent of the type of contraction) or undergoing eccentric contraction (then it can occur at any length condition) (Herzog, 2018).

Hypermobile type

If you have been tested as positive for generalized hypermobility (see Chapters 7 and 19), then we recommend that you perform only the first move of each set over the full available range of motion, and that you continue to conduct all subsequent moves over a shorter range of motion only, close to the shortest length position of the primarily working muscle fibers. For example, in the exercise shown in Figure 43.4, this means loading the knee in a more extended position only. It is also recommended that you do not perform mini-bounces and that you also do not apply the principle “Final exhaustion at end range” with enhanced emphasis. And finally, it is suggested that you conduct the concentric phase—i.e. when the primarily working muscle fibers are actively shortening against resistance—two to three times as slowly compared to the eccentric phases.

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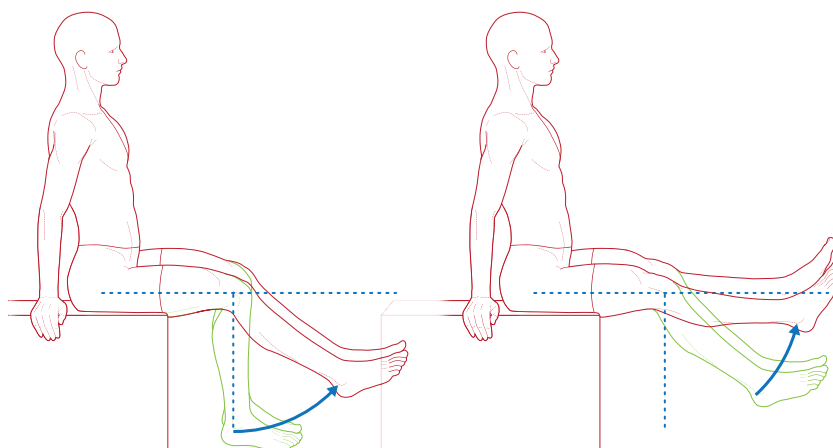


Figure 43.4

Two (of many) different options for conducting knee extension resistance training. In the example on the left, the knee extensors are only loaded in a position at which they are at a longer muscle fiber length. This training style is generally recommended if a long-term increase in knee flexion ability is desired. By contrast, the example on the right shows a resistance training style in which the loaded muscle fibers tend to be at a shorter length. This style is recommended for hypomobile individuals as well as for cancer fighters.

These recommendations are based partly on the new insights about titin signaling, and also on increasing evidence of a long-term shortening effect on muscle fiber length when repeatedly challenging the muscle in a short position against resistance (Gajdosik, 2001). While titin signaling, which we suggest utilizing for hypomobile (Viking) body types, tends to increase muscle fiber length by an increase in the number of the sarcomeres (i.e. the functional contractile units within one fiber), it is assumed that the opposite happens, i.e. a decrease in the number of sarcomeres, when the maximal loading of the muscle tends to happen within its shortest length condition.

Nevertheless, research on hypermobility seems to indicate a tendency for decreased proprioception (Smith et al., 2013), particularly close to the end position of the available range of motion, at which the muscle fibers are at long length. For this purpose, it is essential that the first cycle of movement is conducted over the full available

range of motion and is performed more slowly and with an even higher proprioceptive attention than the subsequent movements within the same set.

Cross-over type

If you were classified as a cross-over type (Chapter 19) and/or if you happen to search for more flexibility in some areas of your body and for more joint stability in other areas, then a mixture of the above recommendations is suggested. For those joints to which you wish to increase your flexibility, then you apply the training style for a Viking-type individual, and for those joints to which you wish to increase joint stability (rather than mobility), we suggest applying the training style described for the hypermobile condition. If you are not certain about a particular body area or exercise regarding your desired preference, then just follow the general Panther training recommendations.

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Tendon training for athletes

If you are an athlete who is doing regular sports not just for fun or health purposes, but rather for a performance orientation within a competition environment, then we suggest applying the Berlin Method of tendon training (Bohm et al., 2014). This method has been shown to induce an increase in both collagen synthesis and tendon resistance over a period of several months, and was larger via this specific training protocol compared with other training types, which involved either more rapid loading or longer lasting tendon stimulations. An increase in tendon resistance is generally regarded as evoking an increased injury protection within an athletic loading environment (Mersmann et al., 2017).

In this protocol, high loads should be applied in five sets of four repetitions with a stress and relaxation duration of ~3 s. Breaks between sets should be 1–2 min (see Figure 43.5). The muscular contraction should be at very high force ($\geq 85\%$ of isometric voluntary force maximum). Due to the length-force relationship of muscle fiber contractions, such high forces are usually only possible around the mid-range position of the available range of motion. For the Achilles tendon contractions, an angle of $\sim 90^\circ$ at the foot joint is

recommended, and for the patellar tendon contractions, a knee joint angle of $\sim 70^\circ$.

Cancer fighters

Resistance training has been shown to reduce mortality in cancer survivors (Hardee et al., 2014). Fascia researcher Thomas Findley advocates a gym-based resistance training, in which the loading happens primarily in the short length position (Findley, 2015). Oriented on a 10-repetition maximum (10RM), patients perform sets of training by performing the first set of 10 at 50% 10RM, the second at 75% 10RM, and the third and final set at 100% of the 10RM. He showed that if the muscles are loaded close to their short length position then the longitudinal stress is less than at mid-range or long positions. However, the lateral stress—forces passing via the epimysium to structures parallel to the working muscle fibers—is much higher (Findley et al., 2015). Findley cites preliminary results which indicate that such a force distribution results in a lowering of TGF- β expression, a substance which is usually regarded as having an aggravating effect on tissue stiffness and cancer progression (Findley, 2015). Figure 43.6 illustrates an example of a preferred loading position in this protocol.

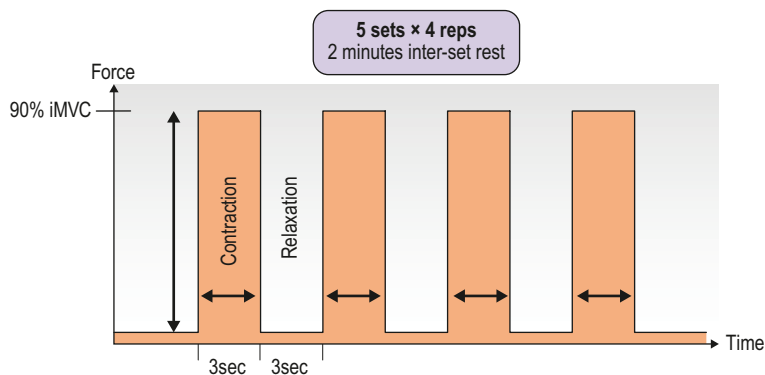


Figure 43.5

Training protocol of the Berlin Method of tendon training for the purpose of enhanced injury (and overload pathology) protection in competitive athletes. High intensity loading to the tendon should be applied in five sets of four repetitions with a contraction and relaxation duration of 3 s each, and an inter-set rest of 2 min.

Illustration from Mersmann et al. (2017).

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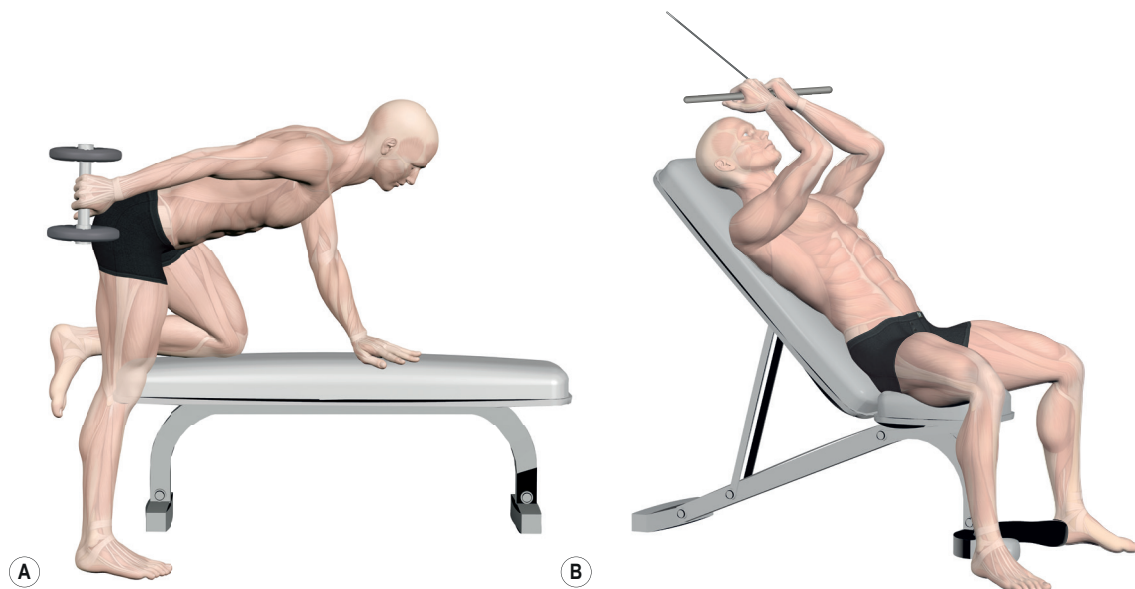


Figure 43.6

Examples of different triceps curls. **(A)** With the muscle fibers at short length, and **(B)** with the muscle fibers at long length. The suggested training protocol for cancer fighters recommends resistance training around the short length position **(A)**.

Clinical summary

Fascia-oriented training does not need to stop at the gym door. Several training recommendations have been developed by different authors for using conventional muscle training machines in a modified manner with the aim of influencing fascial tissue remodeling. Different protocols have been presented for specific mobility

body types as well as for other purposes. Fascia-oriented exercises in a gym environment should be regarded not as an alternative but rather as complementing muscle strength-oriented training. We recommend practicing fascia-oriented training and conventional muscle training on different days with sufficient resting time in between to allow for adequate recovery and adaptation.

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